

JECRC UNIVERSITY JAIPUR
Session: 2025-26
Subject: Mathematics-I (DMA024A)
Course: B. Tech - I Semester
Tutorial Sheet– 12

UNIT-IV

M.M: 64

CO4: To deal with functions of several variables that are essential in most branches of engineering.

PART – A

(5x2=10)

- Q.1 Define a stationary point of a function of two variables.
- Q.2 State the conditions for a point (x_0, y_0) to be a local maximum or local minimum of $f(x, y)$.
- Q.3 Write the formula for the discriminant D used in classifying stationary points of $f(x, y)$.
- Q.4 Define a saddle point with an example.
- Q.5 Write the general form of the Lagrange function for a function $f(x, y)$ subject to the constraint $g(x, y) = 0$.

PART – B

(3x7=21)

- Q.1 Find and classify the stationary points of $f(x, y) = x^2 + y^2 - 2x - 4y + 5$.
- Q.2 Determine whether the function $f(x, y) = x^3 - 3xy^2$ has a saddle point at the origin.
- Q.3 Using the method of Lagrange multipliers, find the maximum and minimum values of $f(x, y) = x^2 + y^2$ subject to $x + y = 4$.

PART – C

(3X11=33)

- Q.1 Find the local maxima, minima, and saddle points of $f(x, y) = x^4 + y^4 - 4xy + 1$.
- Q.2 Using the method of Lagrange multipliers, find the maximum and minimum values of $f(x, y) = xy$ subject to the constraint $x^2 + 4y^2 = 16$.
- Q.3 Find the stationary points of $f(x, y) = x^3 + y^3 - 3xy$ and determine their nature.

Answers: - Part – A

Part – B

1. Strict local minimum on stationary point (1, 2), and Min. value of $f(1, 2) = 0$.
2. The stationary point (0, 0) is a saddle point.
3. Minimum value = 8 at (2, 2), and no maximum.

Part – C

1. Local minimum at (1,1) and (-1, -1), and Saddle point at (0,0).
2. Maximum = 4, at $(2\sqrt{2}, \sqrt{2})$ and $(-2\sqrt{2}, -\sqrt{2})$. Minimum = -4, at $(2\sqrt{2}, -\sqrt{2})$ and $(-2\sqrt{2}, \sqrt{2})$.
3. The stationary points are (0, 0) and (1, 1). Saddle point at (0, 0), and Local Minimum at (1, 1).